

Replication Package for

Do Higher-Cost Settings Deliver Better Outcomes? Post-Acute Care Efficiency in Medicare Skilled Nursing Facilities and Home Health Agencies

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Setup:

Programming language: The codes are written in STATA, and the authors used STATA 17–19 for all the analyses. You will need STATA license to replicate all the files.

Before running the analysis, please install the following user-written packages in STATA:

```
ssc install ftools, replace
ssc install reghdfe, replace
ssc install ivreghdfe, replace
ssc install estout, replace
ssc install ivreg2, replace
net install ddml, from(https://raw.githubusercontent.com/aahrens1/ddml/master) replace
```

Replication folders:

The replication files include four folders. "code," "data," "temp," "output."

- code: config.do, master.do, and one program per table/figure.
- data: restricted analytic files are placed here by approved researchers.
- temp: intermediate files created during execution.
- output/figures: PNG figure outputs.
- output/tables: RTF or XLSX table outputs.

Data access:

The data used in this study include restricted Medicare administrative claims and enrollment files from the Centers for Medicare & Medicaid Services (CMS), including the Medicare Provider Analysis and Review (MedPAR) file, Medicare Beneficiary Summary File (MBSF), and Residential History File. These data were obtained under a CMS Data Use Agreement and cannot be publicly shared because of CMS privacy, confidentiality, and data-use restrictions.

Qualified researchers may apply for access to these data directly through CMS by completing the required application and approval procedures. Researchers seeking additional information about the data-access procedures or data-generation process can contact the authors.

Required analytic files:

File	Sample	Used for
data/analysis_main.dta	One record per eligible SNF/HHA index hospitalization; stroke and CHF cohorts; 2013–2017	Figures 1/A2; Tables 1–3, 5–6, A3–A11
data/analysis_destinations.dta	One record per eligible index hospitalization in the broader destination (SNF, HHA, IRF, LTCH, and home) sample; stroke and CHF cohorts; 2013–2017	Table 4 and Table A1
data/analysis_market_2017.dta	One record per HSA in 2017	Table A2
data/analysis_market_thresholds.dta	One record per eligible SNF/HHA index hospitalization without sample restrictions; stroke and CHF cohorts; 2013–2017	Table A12

Code inventory:

Stata file	Output reproduced
figure 1.do	Occupancy distribution and trends
table 1.do	Balance by high/low occupancy
table 2.do	SNF/HHA descriptive statistics
table 3.do	First stage
table 4.do	Alternative discharge destinations
table 5.do	Baseline OLS and 2SLS
table 6.do	Heterogeneity
figure A2.do	Alternative model specifications
table A1.do	First discharge locations
table A2.do	Market characteristics by occupancy quartile
table A3.do	Timing placebo F-statistics
table A4.do	IV validity checks
table A5.do	Falsification test in low-occupancy markets
table A6.do	Disease-specific 2SLS
table A7.do	SNF patient composition
table A8.do	Short-stay placebo sample
table A9.do	Trim occupancy extremes
table A10.do	Alternative outcomes
table A11.do	CUE-GMM and DML-IV
table A12.do	Alternative market thresholds

Standardized variable dictionary:

Variable	Definition
beneficiary_id	De-identified beneficiary identifier
index_id	Unique index hospitalization identifier
hsa	Hospital Service Area identifier
hsa_quarter	HSA $\times$ calendar-quarter fixed-effect identifier
year	Discharge calendar year
quarter	Discharge calendar quarter
disease	1 = stroke; 2 = CHF
discharge_date	Stata daily hospital-discharge date
state_year	State $\times$ year fixed-effect identifier
month_year	Month $\times$ year fixed-effect identifier
snf	1 = initial SNF discharge; 0 = HHA in main file
hha	HHA destination indicator
irf	IRF destination indicator
ltch	LTCH destination indicator
home	Home/no formal PAC destination indicator
occupancy	Daily HSA short-stay SNF occupancy rate
occupancy_lag365	Occupancy 365 days before discharge
occupancy_lead365	Occupancy 365 days after discharge
occupancy_lag1	Occupancy one day before discharge
occupancy_lead1	Occupancy one day after discharge/admission
market_discharges	Cumulative 2013–2017 HSA SNF/HHA discharges
readmit30	All-cause 30-day readmission indicator
mortality30	All-cause 30-day mortality indicator
pac_spending	Medicare spending for the PAC episode, dollars
readmit90	All-cause 90-day readmission indicator
mortality90	All-cause 90-day mortality indicator
mortality180	All-cause 180-day mortality indicator
hospital_los	Index hospital length of stay, days
quality_rating	Mean SNF/HHA five-star quality rating
nursing_hours	Nursing hours per resident day
age	Age at index discharge
female	Female indicator
white	White race indicator
black	Black race indicator

Variable	Definition
zip_income	ZIP-code median household income
dual_full	Full dual-eligibility indicator
dual_partial	Partial dual-eligibility indicator
ihd	Ischemic heart disease indicator
copd	COPD indicator
diabetes	Diabetes indicator
prostate_cancer	Prostate cancer indicator
lung_cancer	Lung cancer indicator
colorectal_cancer	Colorectal cancer indicator
breast_cancer	Breast cancer indicator
organ_transplant	Organ transplant indicator
hyperlipidemia	Hyperlipidemia indicator
adrd	Alzheimer's disease/related dementias indicator
osteoporosis	Osteoporosis indicator
rheumatoid_arthritis	Rheumatoid arthritis indicator
heart_valve_disorder	Heart valve disorder indicator
pulmonary_circulation_disorder	Pulmonary circulation disorder indicator
hypertension	Hypertension indicator
neurological_disorder	Neurological disorder indicator
chronic_lung_disease	Chronic lung disease indicator
hypothyroidism	Hypothyroidism indicator
renal_failure	Renal failure indicator
tumor	Tumor diagnosis indicator
coagulation_disorder	Coagulation disorder indicator
anemia	Deficiency anemia indicator
weight_loss	Weight-loss indicator
obesity	Obesity indicator
elixhauser_mortality	Elixhauser 30-day mortality index
elixhauser_readmission	Elixhauser 30-day readmission index
ma_penetration	HSA Medicare Advantage penetration
aco_share	HSA share enrolled in an ACO
distance_irf_km	Distance to nearest IRF, kilometers
con_state	State CON-law indicator
rural	Rural residential ZIP indicator
high_income	ZIP income above sample median indicator

Market-file variable dictionary:

Variable	Definition
hsa	Hospital Service Area identifier
occupancy	2017 HSA occupancy rate used to form quartiles
ma_beneficiaries	Mean MA beneficiary count
ma_share	MA beneficiary share
snf_count	SNF count
hha_count	HHA count
hospital_count	Acute-care hospital count
aco_beneficiaries	ACO beneficiary count
aco_share	ACO beneficiary share
rn_lpn_hours	RN + LPN hours per resident day
zip_income	Median household income
sdi	Social Deprivation Index
share_age65	Population share age 65+
snf_beds	Mean beds per SNF
snf_forprofit_share	For-profit SNF share
snf_rating	Mean overall SNF rating
snf_fivestar_share	Five-star SNF share
hha_rating	Mean HHA rating
hha_fivestar_share	Five-star HHA share
hospital_rating	Mean hospital rating